

- 21** An electron of charge  $e$  and mass  $m_e$  enters through a hole into a space between two parallel metal plates AB and CD, separated by a distance  $d$ .

CD is maintained at a positive potential  $V$  relative to AB.



The velocity of the electron is  $v$ .

Which expression is correct for the electron to just reach AB?

- A**  $\frac{1}{2} m_e v^2 = \frac{e}{4\pi\epsilon_0 d}$
- B**  $\frac{1}{2} m_e v^2 = eV$
- C**  $\frac{1}{2} m_e (v \cos \theta)^2 = eV$
- D**  $\frac{1}{2} m_e (v \sin \theta)^2 = eV$

22. A resistor is made of a copper cylindrical core of cross-sectional area  $0.01 \text{ cm}^2$  which